UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES



SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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Submitted To

Al Imtiaz, Associate Professor

Department of CSE

Submitted By

Name: Mitu Barua Omi

ID: 0432320005101032

And

Name: Zakia Sultana Chadni

ID: 0432320005101050

Section: 6A2. Batch: 54

1. Introduction

1.1. Purpose

In recent years, urban gardening and plant enthusiasm have grown rapidly. However, plant care takers often face isolated platforms for buying supplies, diagnosing plant health issues, exchanging excess plants, and obtaining efficient delivery. The BloomVerse platform is designed to integrate all gardening needs into a unified, intelligent web ecosystem. BloomVerse serves as a multi-faceted digital platform that categorizes products into dedicated sections (Plants, Seeds, Fertilizers, Gardening Tools), offers an AI-powered Plant Health Tracker (Doctor) for disease and water issue diagnosis via image upload, facilitates peer-to peer sharing via the Plant Adoption and Community Swap Hub, and optimizes logistics through an intelligent Green Route delivery planner for riders. By bridging commerce, community engagement, artificial intelligence, and eco-friendly delivery logistics, BloomVerse creates a holistic standard for digital gardening solutions.

1.2 Document Conventions

A. The entire document justified.

B. Convention for Main title

- Font face: Times New Roman
- Font style: Bold
- Font Size: 14

C. Convention for Sub title

- Font face: Times New Roman
- Font style: Bold
- Font Size: 14

D. Convention for body

- Font face: Times New Roman
- Font style: Normal
- Font Size: 12

1.3 Scope of Development Project

This system replaces traditional fragmented gardening platforms with an integrated web application built using HTML, CSS, JavaScript, PHP, and MySQL Database. It enables users to:

- Browse and purchase products sorted into distinct sections: Plants, Seeds, Fertilizers, and Gardening Tools.
- Upload plant photos to analyze and detect diseases or watering deficiencies using the Plant Health Tracker (AI Doctor).
- Donate or swap old plants with community members free of cost via the Plant Adoption and Community Swap Hub.
- Utilize the Green Route feature which plans the shortest, time-optimized routes for delivery riders.
- Manage user authentication, profile dashboards, cart management, and admin supervision.

1.4 Definitions, Acronyms and Abbreviations

- Admin: System administrator with full control over products, users, orders, swap moderation, delivery logs, and reports.
- User / Customer: Registered plant enthusiast who uses the shopping, swapping, and diagnostic features.
- Delivery Rider: Field agent who uses the Green Route module to deliver orders.
- AI Doctor / Plant Health Scanner: Rule-based or machine learning image diagnostic feature.
- DBMS: Database Management System (MySQL).
- Figma: A collaborative design tool used to create the UI/UX prototype of BloomVerse.
- SRS: Software Requirements Specification.

- Eco-Certified: A product label for items made from biological or organic compounds, verified by the admin.
- Moderation Queue: Admin list of flagged swap listings and reviews waiting for action.

2. Overall Descriptions

2.1 Product Perspective

BloomVerse is a web-based client-server platform designed using native core web technologies (HTML5, CSS3, JavaScript, PHP, MySQL Database). It consolidates commercial, diagnostic, social, and logistical features under a cohesive web system. From a technical architecture standpoint:

- Frontend: Dynamic user interfaces engineered with HTML, CSS, and JavaScript for seamless responsiveness.
- Backend: Modular server-side processing using PHP to manage authentication, business rules, plant health logic, and routing algorithms.
- Central Database: MySQL relational DBMS storing user credentials, catalog products, swap listings, diagnostic logs, and routing data.

2.2 Product Function

The major system functions of BloomVerse include:

1. User Registration & Authentication: Secure sign-up, login, password encryption (bcrypt/ md5), and role-based access control (Admin, User, Rider).
2. Categorized E-Commerce Storefront: Dedicated browsing sections for Plants, Seeds, Fertilizers, and Gardening Tools with search, filter, and cart functionality.
3. Plant Health Tracker (AI Doctor): Image upload tool analyzing plant leaf photos to detect fungal/bacterial diseases or watering issues and recommend solutions.
4. Plant Adoption & Community Swap Hub: Platform enabling users to list existing plants for free adoption or swap with other members.

5. Green Route Delivery Planner: Dedicated rider dashboard calculating shortest paths and minimum travel time routes for eco-friendly dispatches.
6. Shopping Cart & Checkout: Session-managed cart, order placement, and receipt generation.
7. Admin Control Panel: Complete management dashboard for inventory, user moderation, swap requests, and system diagnostics.
8. Database Management: Secure data persistence and relational integrity via MySQL DBMS.

2.3 User Classes and Characteristics

2.3.1. Administrator (Admin):

Characteristics: High technical knowledge, full administrative control. Responsibilities: Manage product catalogs, verify swap posts, monitor Green Route logs, and oversee user activity.

2.3.2. User / Customer / Plant Lover:

Characteristics: Basic to moderate web literacy. Responsibilities: Purchase products, upload plant photos for health checkup, list/adopt plants in community hub, manage orders.

2.3.3. Delivery Rider:

Characteristics: Mobile/web user navigating delivery routes. Responsibilities: Access the Green Route dashboard, view optimal paths, update package delivery status.

2.3.4. System Perspective on Users:

Access is restricted via PHP sessions and role-based access control ensuring users only access authorized modules.

2.4 Operating Environment

BloomVerse operates on modern web browsers (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari) across desktops, laptops, tablets, and smartphones.

2.5 Assumptions and Dependencies

Assumptions:

- Users have stable internet access and camera-enabled devices for uploading plant pictures.
- Riders use internet-connected smartphones for viewing route paths.

Dependencies:

• Active MySQL Database connection and Web Server hosting.

2.6 Requirement

Functional Requirements:

- User login, signup, and profile management.
- Sectional product viewing (Plants, Seeds, Fertilizers, Tools).
- Plant image upload for health diagnostic checks.
- Listing, viewing, and applying for plant swaps and free adoption.
- Shortest path route generation for delivery riders.

Non-Functional Requirements:

- Fast page response.
- Secure password handling and SQL injection prevention.
- Intuitive, aesthetic UI design tailored for gardening enthusiasts.

Hardware & Software Requirements:

- Frontend: HTML, CSS, JavaScript
- Backend: PHP
- Database: MySQL
- Development Tools: XAMPP environment

2.7 Data Requirement

System inputs include registration data, product searches, plant images for health analysis, swap post details, and delivery address coordinates. Outputs include

health diagnostic reports, categorized product pages, swap confirmation notifications, and optimized Green Route maps.

3. External Interface Requirement

User Interface: Clean responsive layouts with separate dashboards for Admin, User, and Rider.

Software Interface: Web browser runtime environment integrated with the PHP engine and MySQL DBMS. A map service or library is used for route display.

Communication Interface: HTTP/HTTPS protocol using standard TCP/IP networking.

4. System Features

- Home Page: Hero banner with Explore Now and Learn How It Works buttons, platform statistics, Marketplace category cards, AI Doctor preview, Swap Hub preview, Green Deliveries tracker, testimonials, and footer.
- Categorized Shop: Filterable product grid with category, price range, rating, and Eco-Certified filters, sorting, and pagination.
- Reviews and Ratings: Star ratings and written reviews on products, with a flag option.
- AI Plant Health Scanner: Image drop zone, health index ring, issue cards with severity, organic remedies with shop links, and scan history.
- Community Swap Hub: Listing cards with type badge, distance, and owner, a Create Listing form, and an Interested action.
- Green Route Planner: Rider map, active route orders with status, shift toggle, and daily statistics.
- User Authentication: Login, registration, and password reset.
- Cart and Checkout: Session-based cart, order submission, and receipt.
- Admin Dashboard: Statistics cards, recent orders, moderation queue, and a sidebar for all management modules.

- Responsive Mobile Design: Adaptive CSS grid and flexbox for smooth smartphone viewing.

5. Other Non-functional Requirements

5.1 Performance Requirement

The system must support concurrent user interactions without performance lag and maintain low database query response times.

5.2 Safety Requirement

All user uploads must be sanitized to prevent file execution vulnerabilities on the server.

5.3 Security Requirement

Input validation, SQL injection protection via PHP.

5.4 Requirement attributes

Each requirement contains Unique ID, Type (Functional/Non-Functional), Priority, Source, and Implementation Status.

5.5 Business Rules

- Users must be registered and logged in to post in the Swap Hub or order products.
- Free adoption items cannot have price values attached.
- Delivery riders can see route calculations only for orders assigned to them.
- Only customers who received a product can review it.
- Only admins can mark a product as Eco-Certified.

5.6 User Requirement

Users require a visually pleasing, easy-to-navigate web portal to meet all their plant care, commerce, community, and delivery expectations.

6. Other Requirements

6.1 Data and Category Requirement

MySQL tables: Users, Roles, Categories (Plants, Seeds, Fertilizers, Tools), Products, Reviews, Cart_Items, Orders, Order_Items, Health_Logs, Swap_Posts, Swap_Interests, Route_Details, and Moderation_Reports.

6.2 Glossary

BloomVerse: The name of the integrated software platform.

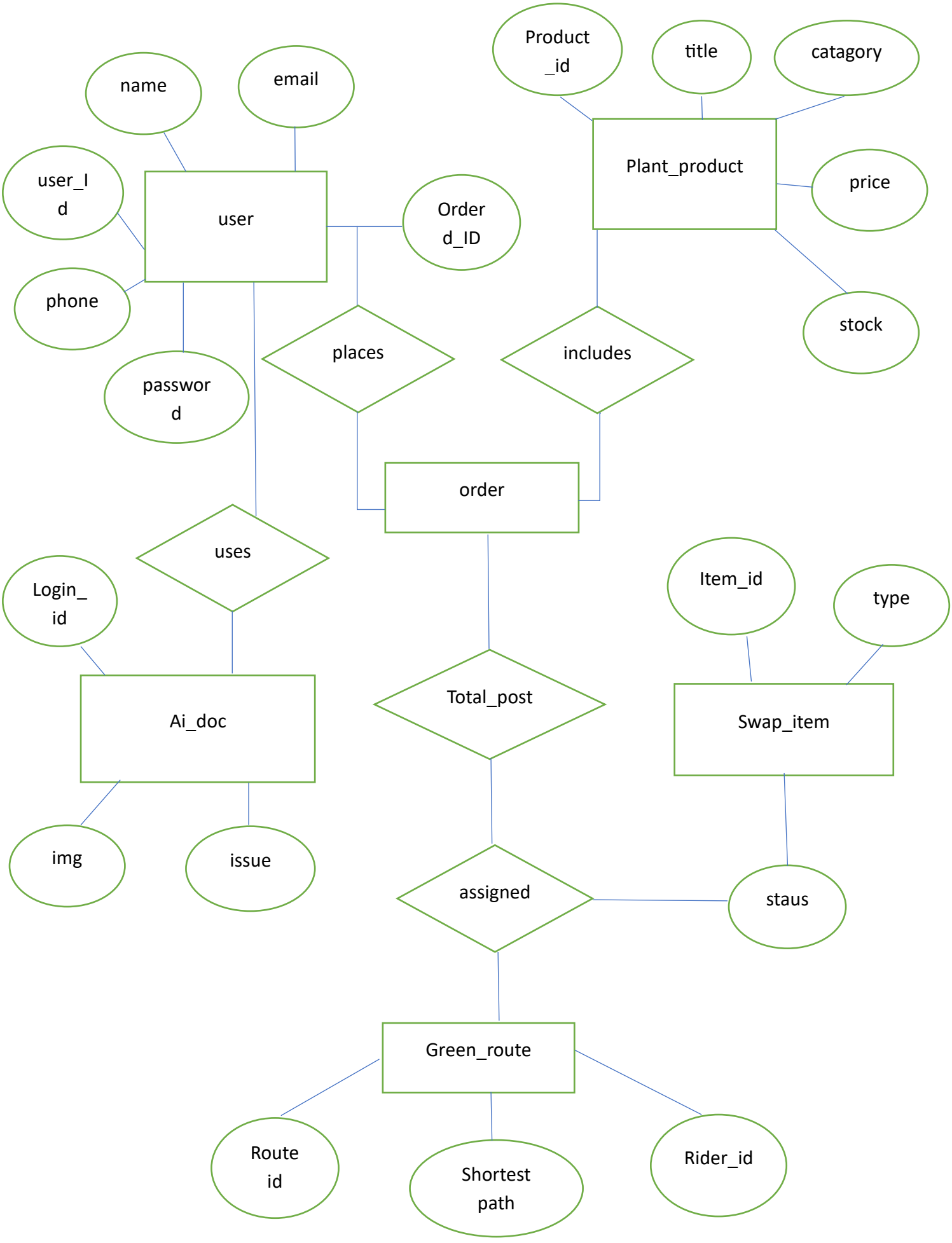
Green Route: An optimized path-finding module for eco-friendly and fast deliveries.

Swap Hub: A community portal for free plant adoption and peer trading.

Health Index: Percentage score that shows the overall health of a scanned plant.

6.3 ER Diagram

An entity is a user-defined description of a real-world object or concept. It identifies the attributes of the data and the unique keys that distinguish instances of the data. An entity of data has a name, a set of attributes that describes its characteristics, and a set of relationships that connect it to other entities. In the Bloomverse system, the primary entities include User, Product, Plant Health Tracker, Swap & Adoption Hub, and Order & Green Route. The structural entities and their cardinalities are illustrated in the diagram below:


Product _id
title
catagory
name
email
Plant_product
user_I d
price
user
Order d_ID
phone
stock
places
includes
passwor d
order
uses
Login_ id
Item_id
type
Total_post
Ai_doc
Swap_item
img
issue
assigned
staus
Green_route
Route id
Shortest path
Rider_id

7. Conclusion

The BloomVerse platform is a complete software system that unites e-commerce, artificial intelligence health diagnostics, peer-to-peer plant sharing, and optimized delivery logistics into a cohesive web architecture. Developed using core web technologies (HTML, CSS, JavaScript, PHP, and MySQL), this project demonstrates how technology can promote eco-friendly practices, sustainable urban gardening, and efficient digital commerce.